End Term Report

SPARK 22

Study of Optical Photons using Geant4 Simulation Toolkit

Submitted by

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Declaration

I, **Utkarsh Bajpai** (SPARK ID: SPA2201209), hereby declare that, this report entitled **Study of Optical Photons using Geant4 Simulation Toolkit** is an original work carried out by me under the supervision of **Prof. Anil Kumar Gourishetty** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold the academic ethics and honesty. Whenever an external information or statement or result is used then, that have been duly acknowledged and cited.

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July 2022

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Abstract

Scintillation detectors are very useful for finding the energy and presence of gamma sources. Optical Photons are generated in a scintillator crystal when gamma radiation falls on it. Study of optical photons, their generation, transport and reflections has been performed using Geant4 Simulation Toolkit. Simulation of scintillation detectors to obtain Pulse Height spectra is achieved. Detector Efficiency using Geant4 has also been estimated. Vidmar Efficiency Calibration method has also been understood.

Keywords: Optical Photon Transport, Scintillators, GEANT4, Detector Efficiency, Intrinsic Resolution, Light collection efficiency

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1 Literature Review

1.1 Novel Probability Distribution Function of Optical Photons

A novel mathematical model for radial scintillation light distribution on the detector surface for the case of continuous scintillation crystal coupled through a light guide to a detector [1] has been formulated.

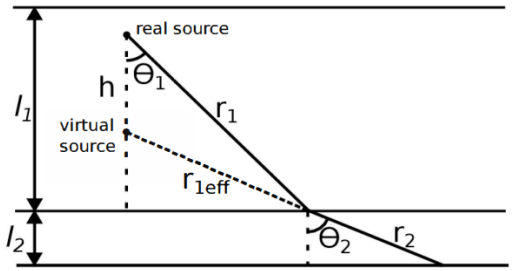


Figure 1: Simplified scheme of refraction effect. Radial position of virtual source is taken as same a s virtual.

The intensity of the scintillation light on the detection surface as a function of the angles of incidence and refraction is given as

$$I(\theta_1, \theta_2) = \alpha \cdot \frac{E_g \gamma}{4\pi}[1 - R(\theta_1, \theta_2)]\frac{\cos\theta_2}{r^2{}_{eff}}$$

Here, γ is crystal light yield. E_g is the energy of incident gamma. α is conversion factor to convert photon number into light power.

Position corresponding to $I(\theta_1, \theta_2)$ is given as

$$x(\theta_1, \theta_2) = h \tan\theta_1 + l_2 \tan\theta_2$$

Thus, the probability density function is given as

$$P_x(x) = \frac{I(x)}{\int_0^\infty I(x)dx}$$

This model had a computational time four order of magnitude lower than the Monte Carlo simulation for same number of scintillation photons. It agrees well with MC simulations especially for optical photon generation at top of crystal.

1.2 Effect of Scintillator shape on Light Collection Efficiency

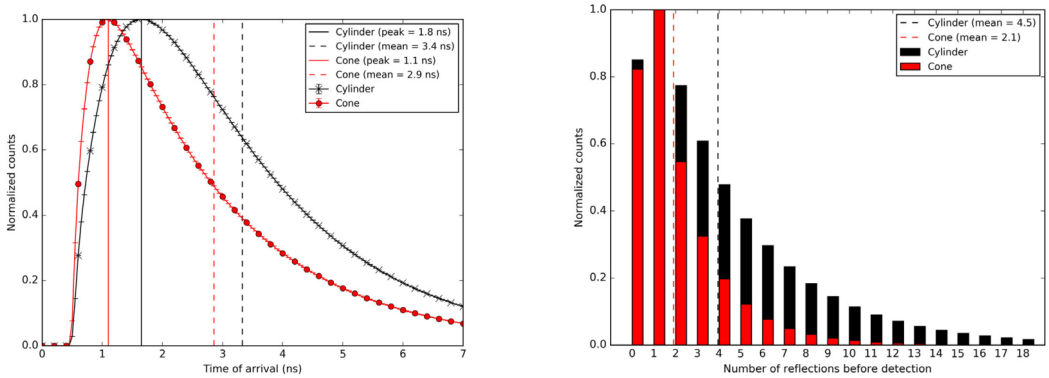
The energy resolution for conical vs cylindrical geometry scintillators were compared [2]. The experimental results indicate that cone has a better light collection efficiency as compared to the cylinder.

This was mainly due to higher number of total internal reflections in cylinder (symmetric), the probability of detection(P_d) reduced with number of reflections (N).

$$P_d \propto (R_t)^N$$

R_t is the total reflectivity coefficient($R_t < 1$).

The study was supported by simulations of the time of arrival (ToA) and number of reflections (NOR) of optical photons cf Figure 2.



(a) Time Of Arrival(ToA)

(b) Number of Reflections (NOR)

Figure 2

1.3 Optimization of Scintillation parameters using simulations

A 2" × 2" CsI(Tl) scintillator was simulated in Geant4 [3]. Gamma photos were transported towards the scintillator crytsal leading to generation and transport of optical photons in the crystal upto the coupled photo-multiplier tube (PMT). A pulse height spectrum was generated from the number of photons produced corresponding to each gamma.

It was compared with experimental spectrum for code validation. The dominant parameters that influence the pulse height spectra were identified, namely, quantum efficiency of the detector, matching of its spectral response with, reflectivity of scintillation coating material and absolute light yield.

These parameters were optimized by varying them and comparing their resolution and pulse height spectra with experiments.

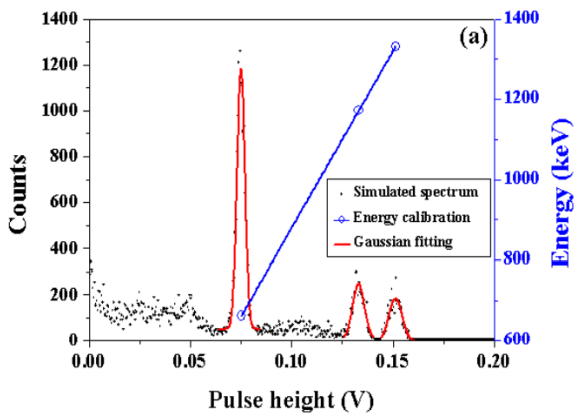


Figure 3: Simulated Pulse Height Spectra for Cs 137 + Co 60 source

1.4 Vidmar Energy Calibration for peak efficiency determination of radio-nuclides with a two-step cascade decay

The Vidmar energy callibration method [4] provides a nice mathematical formalism to determine photo peak efficiency for case of multi energetic spectra like Cobalt 60. Thus, each bi-energetic source can be used to find two efficiency data points. These can be included in the energy calibration curve.

The number of photons $N_1,\ N_2,\ N_{12}$ in the peaks corresponding to energies $E_1,\ E_2$ and E_{12} are related to source activity A, full energy peak efficiencies $\epsilon_1,\ \epsilon_2$ and total efficiencies η_2, η_1 through following equations.

$$\dot{N_1} = A\epsilon_1(1-\eta_2)$$
$$\dot{N_2} = A\epsilon_2(1-\eta_1)$$
$$\dot{N_{12}} = A\epsilon_1\epsilon_2$$

Thus, we have 3 equations and 4 unknowns. Additional physical constants are required for making the system of equation tractable. The following inequalities hold for gamma ray energies above 200 keV.

$$\epsilon_1 > \epsilon_2$$
$$\eta_1 > \eta_2$$
$$\eta_2/\epsilon_2 > \eta_1/\epsilon_1$$

From these inequalities a lower and an upper bound on ϵ_1 has been obtained. Providing that the determined range of values is sufficiently narrow and assuming a uniform probability distribution for ϵ_1 to lie in $[\epsilon_{min}, \epsilon_{max}]$. ϵ_1 can be estimated as

$$\epsilon_1 = [\epsilon_{min} + \epsilon_{max}]/2$$

The other three unknows can now be solved from the three equations.

2 Geant4

Geant4 (for GEometry ANd Tracking) is a simulation package developed by RD44, worldwide collaboration of institutes. It is open source and currently is being maintained by CERN. It is widely used in high energy, nuclear and accelerator physics. The user inputs the geometry, material, properties of the detector; the necessary physics to be invoked; the particle source configuration; quantities to be scored. The output provides us accurate simulations of passage of particles through matter using Monte Carlo methods.

2.1 Installation of Geant4

Geant4, simulation toolkit was installed on a virtual machine. The version used was `Geant4 10.11 patch 02`. The Linux version was `Ubuntu 20.04`. We then played with basic examples to understand the basics of Geant4 like detector geometry, particle generation, physics lists, scoring values of interest.

We used Geant4 Installation Guide as well as Physics Matters Geant4 playlist for installation. The major steps for installation were.

1. Downloading the latest source code from the distribution page.

2. Unpacking the code to desired location, here /home/software/geant4.

```
$ cd  /home/software/geant4
$ tar -xzfv ~/Downloads/geant4.10.11.p02.tar.gz
```

3. Installing Prerequisties and required dependencies for running for Geant4.

```
$ sudo apt install cmake cmake-curses-gui gcc g++ libexpat1-dev qt5-default
  libxmu-dev libmotif-dev
```

4. Configuring the build using ccmake and selecting the installation prefix (folder where Geant4 will be installed), additional build options cf Figure 4. The mandatory options to enable are INSTALL_DATA, QT, RAYTRACER_X11.

```
$ mkdir build
$ cd build
$ ccmake ..
```

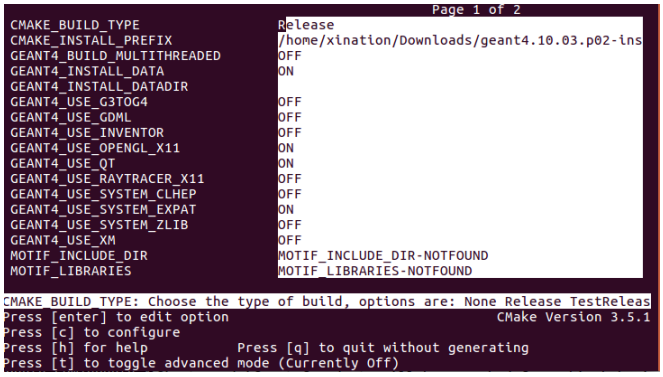


Figure 4: Cmake GUI window

5. Running the make. N is the number of cores

```
$ make -jN
```

6. Installing Geant4 in the directory specified in installation prefix.

```
$ make install
```

Once installation has been completed we shall see four folders have been created.

```
$ ls
$ bin include lib share
```

7. Sourcing the make file. We add the following into the our bash.rc file to set the environment variables.

```
$ source /home/user/geant4/geant4.10.11.p02-install/bin/geant4make.sh
```

8. Running example B1 for testing the installation. cf Figure 5

```
$ cd /path/to/geant4_version name/examples/basic/exampleB1
$ mkdir build && cd build
$ cmake ..
$ make
$ ./ exampleB1
```

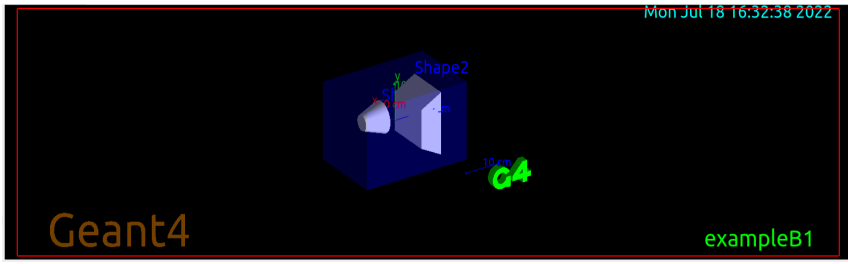


Figure 5: Qt5 visualisation of Example B1 indicating successful installation

2.2 Classes Description

1. `main.cc`: This is the main file that is compiled by make command. Mandatory Initialization classes are set namely, DetectorConstruction.cc, PrimaryGenerator.cc, Physics.cc, Event.cc, Stepping.cc, Run.cc. The visualization manager is also initilaised.

2. `DetectorConstruction.cc`: Contains information about detector geometry, material, material properties, surface definitions and properties, scintilltation parameters like scintilllation yield, resolution scale etc.

3. `PrimaryGenerator.cc`: Contains information about particle type, source position, direction, energy, momentum of primary particle. Additionally, G4GeneralParticleSource provides various different source distributions.

4. `Physics.cc`: Various physics lists to be used in simulation are registered. It depends on the phenomenon being simulated.

5. `Stepping.cc` Consists of the "delta" information about the particle such as energy loss on step, time of flight spent by step etc.

6. `Event.cc` Event is basic unit of simulation in Geant4. It contains results usch as energy deposited from multiple Steps. It passes them to Run.cc.

7. `Run.cc`: Run is a collection of various events that share same detector configuration. It consists the values to be "scored" at end of each event, usually to be entered in a file.

   Apart from the source files, there are corresponding header files that includes the necessary files and also contains initialization of global variables.

2.3 Getting Started

After successfully compiling basic example B1 and others, we started understanding the basic structure of a Geant4 code, how the basic classes are defined for specific purposes,

how the volume are defined using source files of the examples as well as the Geant4 Toolkit for Application Developers.

We played with parameters of example B1 and OpNovice. The geometry shapes, orientation, relative distances were modified and corresponding effects visualised through Qt5 visualisation driver.

Certain slides and lectures were referred for understanding the simulations of optical photons in Geant4 like Optical Photon Processes in GEANT4, Optical Physics, Simulation of Optical Processes in Geant4 Geant4 optical physics.

Example OpNovice2 was studied extensively to understand simulation of scintillation, Cerenkov radiation and scoring the number of optical photons produced. A paper summarizing the various peculiarities and possible solutions in Geant4 Optical Physics was reviewed [5].

Generation of Optical Photons in Geant4 code was understood from G4Scintillation.cc source code. Optical Photons are produced evenly across the track segment involved in the step and emitted uniformly int0 4π solid angle [6]. The photons have random linear polarization perpendicular to the emission direction. Their mean number(N) depend on the parameter SCINTILLATIONYIELD and fluctuations depend on N and RESOLUTIONSCALE.

3 Our Work

3.1 Generation of Optical Photons

After modifying examples B1 and OpNovice we created a cubical box. We then generated a user inputted number of optical photons isotropically from the center. Randomize functions were used to ensure an isotropic distribution of the OP's. Randomize methods were used for generation of optical photons.

```
G4ThreeVector mom(G4UniformRand() - 0.5, G4UniformRand() - 0.5,  G4UniformRand()
    - 0.5);
```

3.2 Reflections of Optical Photons

We reflected the optical generated from five sides of the cube. Thus, the optical photons were focused towards the sixth side cf Figure 9. This phenomenon occurs in a scintillator detector where optical photons generated due to scintillation are focused towards the PMT. Teflon tape wrapped around the crystal is used as reflector. We also placed a sensitive detector to read out the photon positions as they hit at the PMT.

```
//Defining Optical Surface
```

```
G4OpticalSurface* OpReflectorSurface = new G4OpticalSurface("ReflectorSurface",
  unified, polished, dielectric_metal);

//Defining surface reflectivity/efficiency
std::vector<G4double> BoxReflectivity = { 1.0, 1.0 };
std::vector<G4double> BoxEfficiency   = { 0.5, 0.5 };
G4MaterialPropertiesTable* BoxMPT      = new G4MaterialPropertiesTable();
BoxMPT->AddProperty("REFLECTIVITY", ephoton, BoxReflectivity);
BoxMPT->AddProperty("EFFICIENCY", ephoton, BoxEfficiency);
//Attaching reflectivity value to surface
OpBoxSurface->SetMaterialPropertiesTable(BoxPT);

//Attaching logical skin volume to desired volume
G4LogicalSkinSurface* reflector_surface2 = new G4LogicalSkinSurface("
  reflector_surface", logicReflector, OpReflectorSurface);
```

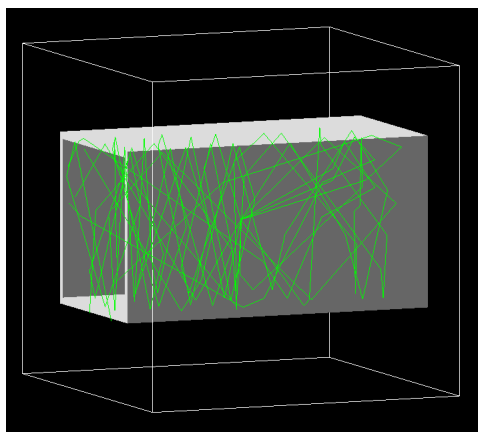


Figure 6: Photons getting Reflected in a 5 sided Al box

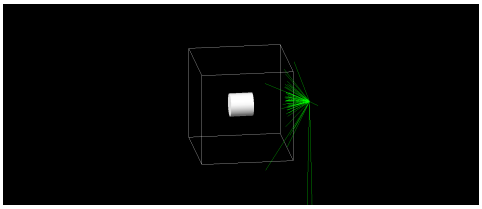
3.3 Shielding Effects

We enclosed the NaI crystal with Aluminum crystal and threw optical photons from outside towards the box. Same exercise without aluminum box was repeated. The OP's were absorbed from the Al surface and unable to penetrate on the box cf Figure 7.

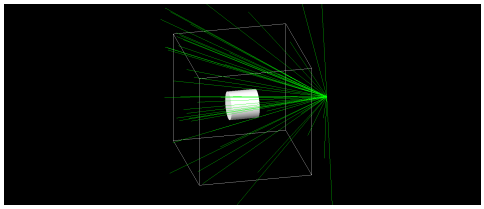
3.4 Effect of Scintillator Shape on the LCE

We tried implementing a paper [2], the main result was that the conical shaped scintillator crystals have a better light collection efficiency(LCE) as compared to cylindrical.

We were able to verify the results by finding the number of Optical Photons hitting the PMT as well as the the time of arrival(ToA) of photons cf Figure 8. For conical the number was 96/100 and ToA was 0.447 ns. For cylindrical 59/100 and 33710 ns.

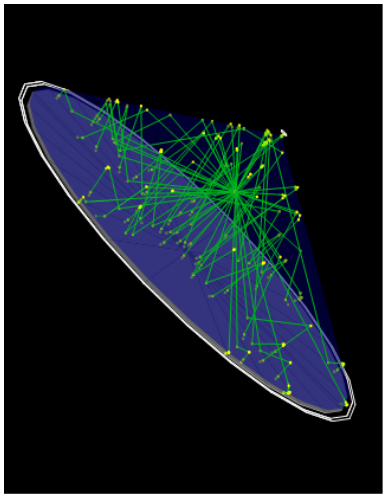


(a) Shielding

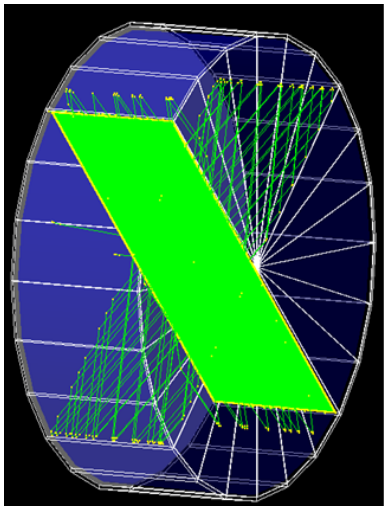


(b) No Shielding

Figure 7: Shielding Effects



(a) Cone



(b) Cylinder

Figure 8: Optical Photons for differnt shapes of Scintillators

Moreover, a continuous profile of scintillation shape vs the number of optical photons reaching the detector was obtained. The plot is shown in Figure 9 .

3.5 Simulation of Scintillation

We modified example OpNovice to simulate scintillator detector for gamma ray detection was simulated using Geant4. It consists of a 2 × 2" CsI crystal. An gamma source was kept at 30 cm from the crystal axis cf Figure 10. The crystal was surrounded by a reflector to trap all the scintillation light. The material density was 4.51 g/cm^3. The `G4OpticalPhysicsClass` incorporates all the optical physics production processes like Scintillation Process, Cherenkov Process and Transition Radiation. As well as the processes at Optical Wavelengths, Refraction and Reflection at medium boundaries, Bulk Absorption, Rayleigh Scattering.

The relevant properties that were set are,

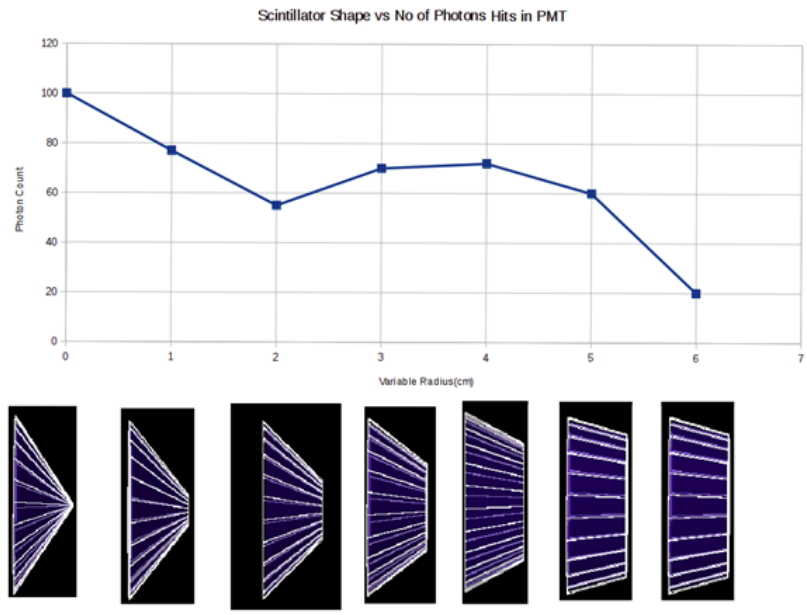


Figure 9: Scintillator Shape vs Photon Number at PMT

1. Photon Energy

```
std::vector<G4double> photonEnergy = {
   2.034 * eV, 2.068 * eV, 2.103 * eV, 2.139 * eV, 2.177 * eV, 2.216 * eV,
   2.256 * eV, 2.298 * eV, 2.341 * eV, 2.386 * eV, 2.433 * eV, 2.481 * eV,
   2.532 * eV, 2.585 * eV, 2.640 * eV, 2.697 * eV, 2.757 * eV, 2.820 * eV,
   2.885 * eV, 2.954 * eV, 3.026 * eV, 3.102 * eV, 3.181 * eV, 3.265 * eV,
   3.353 * eV, 3.446 * eV, 3.545 * eV, 3.649 * eV, 3.760 * eV, 3.877 * eV,
   4.002 * eV, 4.136 * eV
 };
```

2. Absorbtion Length: It is the mean distance travelled by the optical photons before being bulk absorbed (killed).

```
   myMPT1->AddProperty("ABSLENGTH", photonEnergy, absorption);
```

3. Refractive Index : Refractive Index needs to be specified for all energy keyvalues, for propagation of optical photons through that media.

```
   myMPT1->AddProperty("RINDEX", photonEnergy, refractiveIndex1);
```

4. Slow/Fast Scintillation Component: The normalised emission spectra as a function of optical photon energy needs to be converted into fast and slow components according to the ratio $Q1/Q2$ [7].

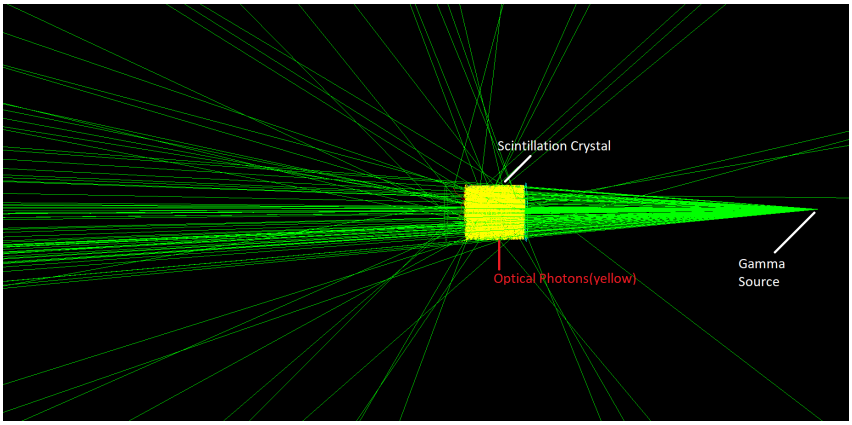


Figure 10: Visualisation of a scintillation event in Geant4

```
myMPT1->AddProperty("SCINTILLATIONCOMPONENT1", photonEnergy, scintilFast);
myMPT1->AddProperty("SCINTILLATIONCOMPONENT2", photonEnergy, scintilSlow);
```

5. Scintillation Yield(LY) : The mean number of optical photons generated in a step is given by $N = LY \times E_{dep}$. Where E_{dep} is the energy deposited in the crystal in that step.

```
myMPT1->AddConstProperty("SCINTILLATIONYIELD", 65000. / MeV);
```

6. Birks Constant: The adjustable parameter in the Birk's formula. It takes into account the non-proportionality of scintillation yield due to quenching effects.

```
CsI->GetIonisation()->SetBirksConstant(0.126 * mm / MeV);
```

7. Resolution Scale: Intrinsic resolution which broadens the statistical distribution of the number of optical photons generated. The optical photons are produced in a Gaussian distribution, with mean N and $\sigma = ResolutionScale \times \sqrt{N}$.

```
myMPT1->AddConstProperty("RESOLUTIONSCALE", 1.0);
```

8. Scintillation Time Constants: Fast and Slow exponential decay time constants of the scintillator crystal.

```
myMPT1->AddConstProperty("SCINTILLATIONTIMECONSTANT1", 700. * ns);
myMPT1->AddConstProperty("SCINTILLATIONTIMECONSTANT2", 3500. * ns);
```

The number of scintillation photons produced were find using the process name in the file `OpNoviceStackingAction.cc`.

```
if(aTrack->GetCreatorProcess()->GetProcessName() == "Scintillation")
      ++fScintillationCounter;
```

For each event the total number of scintillation photons produced were written out in a text file. This was coded in file `OpNoviceStackingAction.cc`.

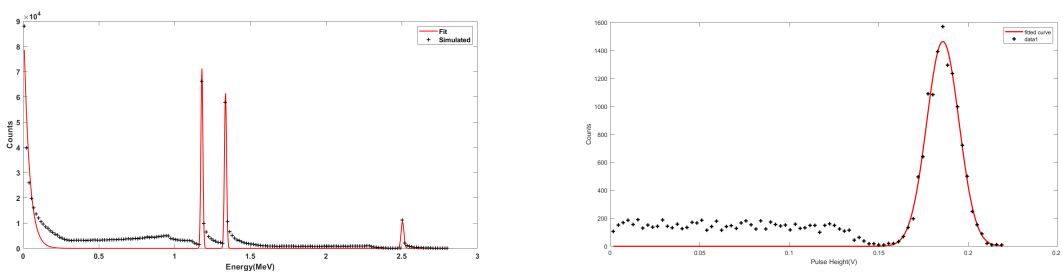
```
ofstream outfile;
  if(fScintillationCounter != 0.0) {
      outfile.open("PhotonCountsPerEvent.txt", ios::out | ios::app );   // open a
    file in write mode.
  }
  if(outfile.is_open()) {
      outfile << fScintillationCounter <<  endl;
      outfile.close(); // close the opened file.
  }
```

At the end of N runs we had a text file having N entries corresponding to number of scintillation photons (n_{ph}) produced in each event(gamma photon hitting the crystal). n_{ph} waa converted into pulse height spectra by using the formula given in the thesis [7].

$$\text{Pulse Height(V)} = \frac{n_{ph} \cdot e \cdot G}{C}$$

Here, e is electronic charge, G is gain of PMT (2.7×10^5 for the Hamamatsu R1306 PMT), C is overall capacitance of the circuit(1 nm, arbitrary).

The generated pulse height data was analysed in MATLAB to obtain the spectra for Cs137 and Co60 respectively cf Figure 11. The presence of sum peak due to true coincidence in case of Co60 shows that G4Ion for C060 works well.



(a) Co 60 (b) Cs 137

Figure 11: Simulated Spectra

3.6 Detector Efficiency

The setup consists of a cylindrical scintillator crytsal of dimensions 3.5 $\phi \times$ 6 L cf Figure 12. A isotropic gamma source was kept at a distance of 1 mm from the crystal. Total 10^6 gamma were generated. The same exercise was repeated for a 1mm Al plate between the crystal and source. Optical Photon generation was suppressed to save computational time (2 hours for 50,000 gamma), working on a virtual machine. Energy deposited by each gamma was

stored. Photopeak(PPE) and Total Efficiency(TE) values were determined from the counts under photo peak and total counts respectively.

For the case of Aluminum, scintillator crystal was defined as a scoring volume and only energy deposited in scoring volume was considered in Stepping.cc class.

```
G4LogicalVolume *fScoringVolume = detectorConstruction->GetScoringVolume();

//If current volume is differnt from scoring volume, we return
if(volume != fScoringVolume)
  return;
//If current volume is fscoring volume the energy deposited in step is added
G4double edep = step->GetTotalEnergyDeposit();
```

The efficiency values are summarised in table below.

Plate	PPE %	TE %
No	22.9	37.19
Yes	19.5	34.25

For the case of Co60 the following results were obtained

Energy(MeV)	Number of Photons
1.173	66000
1.332	57600
2.5	11000

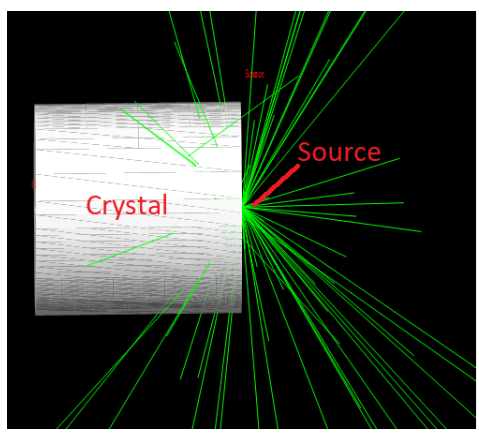


Figure 12: Setup for Efficiency Determination of Detector

4 Future Work

- **Sum Peak Efficiency Determination:** For two-step decay radio isotopes we try to find the efficiency of the sum peak from the area under sum peak. A separate

mathematical formalism needs to be developed for obtaining efficiency of the sum peak. Thus, each single multi energetic source will provide us with three points for efficiency calibration curve.

- **Medical Imaging:** Extending the simulations of optical photons for applications in medical imaging. For this case, single photo-detectors will be replaced by a pixelated array of photodetectors. Spectra from each pixel detector will be used for imaging.
- **Optimisation of Scintillator Parameters:** Simulation of inorganic scintillators (other than CsI) and generation of pulse height spectra from the number of scintillation photons produced. Optical, physical and surface properties can be optimised for new and more complicated settings, without resorting to experiments.

References

[1] M Galasso, A Fabbri, V Cencelli, and L Colace. Propagation of scintillation light in continuous crystals. In *2014 Fotonica AEIT Italian Conference on Photonics Technologies*, pages 1–4. IEEE, 2014.

[2] C.S. Sosa, S.J. Thompson, D.L. Chichester, S.D. Clarke, A. Di Fulvio, and S.A. Pozzi. Energy resolution experiments of conical organic scintillators and a comparison with geant4 simulations. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 898:77–84, 2018.

[3] Pratip Mitra, Mohit Tyagi, R. G. Thomas, A. Vinod Kumar, and S. C. Gadkari. Optimization of parameters for a csi(tl) scintillator detector using geant4-based monte carlo simulation including optical photon transport. *IEEE Transactions on Nuclear Science*, 66(7):1870–1878, 2019.

[4] T. Vidmar, M. Korun, and A. Likar. Close-geometry efficiency calibration in gamma-ray spectrometry using radio-nuclides with a two-step cascade decay. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 508(3):404–413, 2003.

[5] Erik Dietz-Laursonn. Peculiarities in the simulation of optical physics with geant4, 2016.

[6] Geant4 User Documentation. https://geant4.web.cern.ch/support/user_documentation.

[7] Pratip Mitra. *Simulation Based Optimisation Of Indigenously Developed Inorganic Scintillators And Various Light Sensors To Develop Gamma Spectrometer Systems*. PhD thesis, Homi Bhabha National Institute, 2021.